Qassim University | جامعة القصيم
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Information Technology Department

INVESTIGATING THE IMPLEMENTATION OF ARTIFICIAL INTELLIGENCE TECHNOLOGIES IN GOVERNANCE: A RISK FRAMEWORK

Master's Thesis

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1 INTRODUCTION & PROBLEM

AI adoption is accelerating across organizations, but governance practices remain inconsistent, creating risks to accountability, fairness, and cybersecurity.

RESEARCH GAP

Limited empirical frameworks integrate governance principles with cybersecurity to mitigate AI risks.

AI Adoption – Rapid growth across sectors → **Governance Challenges** – Fragmented, inconsistent practices → **Risks & Impact** – Bias, security threats, lack of accountability

AIM
To develop and empirically validate a risk governance framework that enhances the security and resilience of AI technologies in organizational environments

2 RESEARCH OBJECTIVES

- Identify the key gaps in AI governance literature.
- Determine the core components of AI security governance.
- Develop a risk governance framework for AI technologies.
- Empirically test the proposed model.
- Assess the explanatory and predictive importance of the studied dimensions.

3 PROPOSED RISK GOVERNANCE FRAMEWORK

The framework integrates governance principles to enhance AI security and organizational resilience.

Governance Integration
- Policies
- Standards
- Compliance
- Transparency

Bias Mitigation & Fairness
- Fairness
- Equity
- Bias Control
- Ethical AI

Accountability
- Responsibility
- Auditability
- Explainability
- Oversight

Security & Resilience
- Cybersecurity
- Privacy
- Robustness
- Risk Mitigation

→ Significant Relationship - - → Non-significant Relationship

4 METHODOLOGY

Systematic Literature Review (SLR) →
PRISMA-guided review
3 databases
2918 studies
→ 95 selected

Identification	
Identlication	2918
Screening	817
Elipbility	259
Included	95

PRISMA Flow

Survey Design & Data Collection →
Bilingual questionnaire (Arabic & English)
Target: IT professionals in Saudi Arabia

149
Valid Responses

Data Analysis PLS-SEM →
Model evaluation using SmartPLS 4
- Measurement Model
- Structural Model
- Hypothesis Testing

SmartPLS

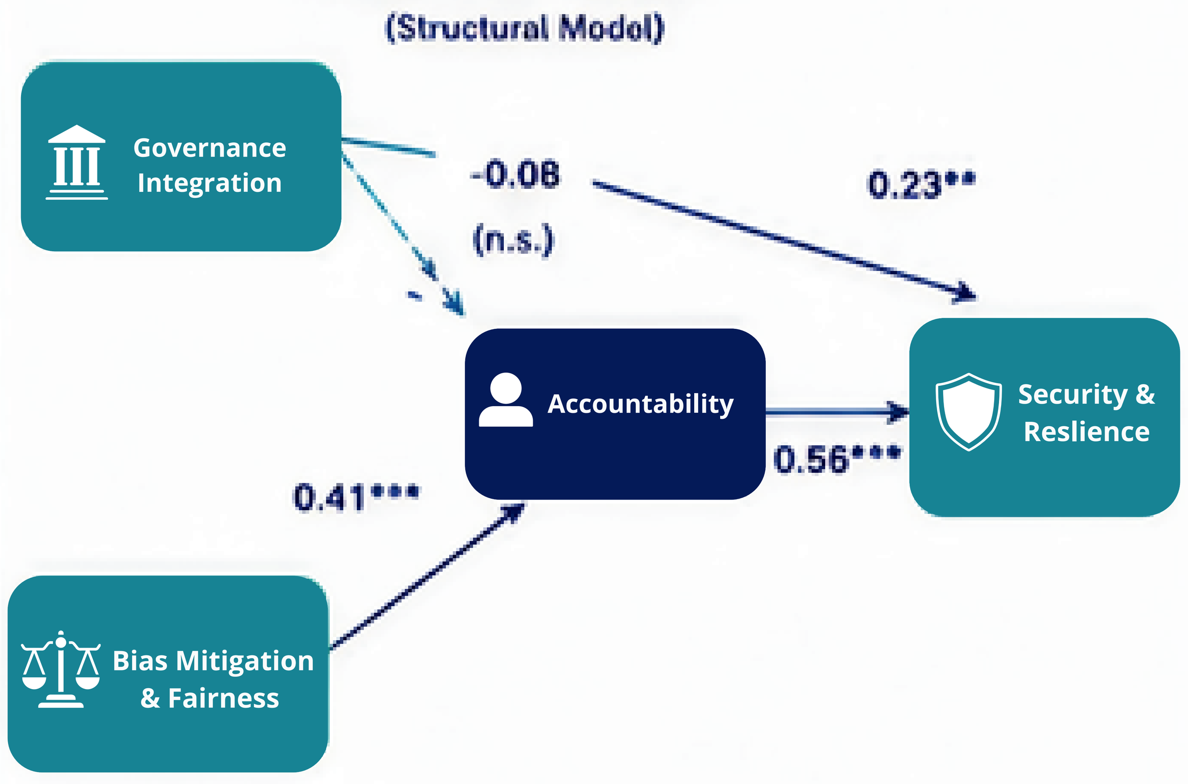
Data Analysis ANN →
Artificial Neural Network using SPSS
- Model Training
- Importance Analysis
- Predictive Power

SPSS

Validation & Evaluation
Assess explanatory and predictive importance of constructs

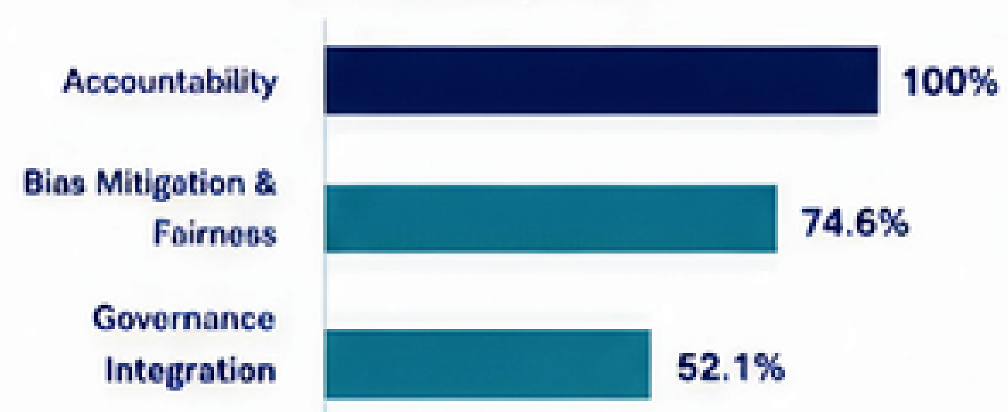
5 RESULTS

PLS-SEM PATH RESULTS
(Structural Model)



*** p < 0.001 ** p < 0.01 * p < 0.05 n.s. = not significant

ANN – PREDICTIVE IMPORTANCE
(Normalized Importance)



KEY INSIGHTS
- Governance Integration did not significantly influence Accountability.
- Accountability is the strongest predictor of Security & Resilience.
- All three constructs significantly enhance Security & Resilience

6 KEY CONTRIBUTIONS

THEORETICAL CONTRIBUTIONS
- Integrates governance and cybersecurity perspectives for AI.
- Provides an empirically validated risk governance framework.
- Advances knowledge on predictors of AI security and resilience.

PRACTICAL CONTRIBUTIONS
- Offers actionable governance guidelines for organizations.
- Supports policymakers in building measurable and auditable governance controls.
- Enhances organizational readiness for responsible AI adoption.

CYBERSECURITY PERSPECTIVE
- Addresses AI-specific security risks.
- Strengthens resilience, privacy, and ethical AI practices.
- Aligns AI governance with cybersecurity best practices

7 FUTURE WOEK

> Expand validation across different sectors and countries.
> Examine long-term governance maturity and evolution.
> Investigate additional AI governance dimensions (e.g., data quality, privacy by design).
> Develop smart governance tools for real-time monitoring.

Koywords
AI Governance, Accountability, Security & Resilience, Bias Mitigation & Fairness, Governance Integration, Cybersecurity